7. Questions & Answers on Counters

Counters - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Counters”.

1. In digital logic, a counter is a device which
a) Counts the number of outputs
b) Stores the number of times a particular event or process has occurred
c) Stores the number of times a clock pulse rises and falls
d) None of the Mentioned
View Answer

Answer: b
Explanation: In digital logic and computing, a counter is a device which stores (and sometimes displays) the number of times a particular event or process has occurred, often in relationship to a clock signal.

2. A counter circuit is usually constructed of
a) A number of latches connected in cascade form
b) A number of NAND gates connected in cascade form
c) A number of flip-flops connected in cascade
d) None of the Mentioned
View Answer

Answer: c
Explanation: A counter circuit is usually constructed of a number of flip-flops connected in cascade.

3. What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of ‘n’ number of flip-flops?
a) 0 to 2n
b) 0 to 2n-1
c) 0 to 2n+1
d) 0 to 2n+1/2
View Answer

Answer: c
Explanation: The maximum possible range of bit-count specifically in n-bit binary counter consisting of ‘n’ number of flip-flops is 0 to 2n+1.

4. How many types of counter are?
a) 2

b) 3
c) 4
d) 5
View Answer

Answer: b
Explanation: Counters are of 3 types, namely, (i)asynchronous/synchronous, (ii)single and multi-mode & (iii)modulus counter.

5. A decimal counter has _____ states.
a) 5
b) 10
c) 15
d) 20
View Answer

Answer: b
Explanation: Decimal counter is also known as 10 stage counter. So, it has 10 states.

6. Ripple counters are also called
a) SSI counters
b) Asynchronous counters
c) Synchronous counters
d) VLSI counters
View Answer

Answer: b
Explanation: Ripple counters are also called asynchronous counter.

7. Synchronous counter is a type of
a) SSI counters
b) LSI counters
c) MSI counters
d) VLSI counters
View Answer

Answer: c
Explanation: Medium Scale Integrated (MSI) is an operation in which the clock pulse is supplied to all the flip-flops simultaneously.

8. Three decade counter would have
a) 2 BCD counters
b) 3 BCD counters
c) 4 BCD counters
d) 5 BCD counters
View Answer

Answer: b
Explanation: Three decade counter has 30 states and a BCD counter has 10 states. So, it would require 3 BCD counters.

9. BCD counter is also known as
a) Parallel counter
b) Decade counter
c) Synchronous counter
d) VLSI counter
View Answer

Answer: b
Explanation: BCD counter is also known as decade counter because both have the same number of stages.

10. The parallel outputs of a counter circuit represent the
a) Parallel data word
b) Clock frequency
c) Counter modulus
d) Clock count
View Answer

Answer: d
Explanation: The parallel outputs of a counter circuit represent the clock count.

Asynchronous Counter - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Asynchronous Counter”.

1. How many natural states will there be in a 4-bit ripple counter?
a) 4
b) 8
c) 16
d) 32
View Answer

Answer: c
Explanation: $2^4 = 16$ states.

2. A ripple counter’s speed is limited by the propagation delay of:
a) Each flip-flop
b) All flip-flops and gates

c) The flip-flops only with gates
d) Only circuit gates
View Answer

Answer: a
Explanation: A ripple counter is something that is derived by other flip-flops. Its like a series of Flip Flops. Output of one FF becomes the input of the next. Because ripple counter is composed of FF only and no gates are there other than FF, so only propagation delay of FF will be taken into account.

3. One of the major drawbacks to the use of asynchronous counters is that:
a) Low-frequency applications are limited because of internal propagation delays
b) High-frequency applications are limited because of internal propagation delays
c) Asynchronous counters do not have major drawbacks and are suitable for use in high- and low-frequency counting applications
d) Asynchronous counters do not have propagation delays, which limits their use in high-frequency applications
View Answer

Answer: b
Explanation: One of the major drawbacks to the use of asynchronous counters is that High-frequency applications are limited because of internal propagation delays.

4. Internal propagation delay of asynchronous counter is removed by
a) Ripple counter
b) Ring counter
c) Modulus counter
d) Synchronous counter
View Answer

Answer: d
Explanation: Internal propagation delay of asynchronous counter is removed by synchronous counter because clock input is given to each flip-flop individually in synchronous counter.

5. What happens to the parallel output word in an asynchronous binary down counter whenever a clock pulse occurs?
a) The output increases by 1
b) The output decreases by 1
c) The output word increases by 2
d) The output word decreases by 2
View Answer

Answer: a
Explanation: In an asynchronous counter, the output of 1st flip-flop is given to second flip-flop as clock input. So, in case of binary down counter the output word decreases by 1.

6. How many flip-flops are required to construct a decade counter?
a) 4
b) 8
c) 5
d) 10
View Answer

Answer: a
Explanation: Number of flip-flop required is calculated by this formula: 2^(n-1)<=N<=2^n. 2^4=16and2^3=8, therefore, 4 flip flops needed.

7. The terminal count of a typical modulus-10 binary counter is
a) 0000
b) 1010
c) 1001
d) 1111
View Answer

Answer: c
Explanation: Modulus-10 means count from 0 to 9. So, terminal count is 9 (1001).

8. How many different states does a 3-bit asynchronous counter have?
a) 2
b) 4
c) 8
d) 16
View Answer

Answer: c
Explanation: 2^3=8, so 8 states a 3-bit asynchronous counter have.

9. A 5-bit asynchronous binary counter is made up of five flip-flops, each with a 12 ns propagation delay. The total propagation delay (tp(total)) is
a) 12 ms
b) 24 ns
c) 48 ns
d) 60 ns
View Answer

Answer: d
Explanation: Each bit has propagation delay = 12ns. So, 5 bits = 12ns * 5 = 60ns.

10. An asynchronous 4-bit binary down counter changes from count 2 to count 3. How many transitional states are required?
a) 1
b) 2

c) 8
d) 15
View Answer

Answer: d
Explanation: Transitional state is given by $2^4 – 1 = 15$. So, total transitional states are 15.

11. A 4-bit ripple counter consists of flip-flops, which each have a propagation delay from clock to Q output of 15 ns. For the counter to recycle from 1111 to 0000, it takes a total of
a) 15 ns
b) 30 ns
c) 45 ns
d) 60 ns
View Answer

Answer: d
Explanation: One bit change is 15 ns, so 4-bit change = 15 * 4 = 60.

12. Three cascaded decade counters will divide the input frequency by
a) 10
b) 20
c) 100
d) 1000
View Answer

Answer: d
Explanation: Decade counter has 10 states. So, three decade counters are cascaded i.e. 10*10*10=1000 states.

13. A ripple counter’s speed is limited by the propagation delay of:
a) Each flip-flop
b) All flip-flops and gates
c) The flip-flops only with gates
d) Only circuit gates
View Answer

Answer: a
Explanation: A ripple counter’s speed is limited by the propagation delay of each flip-flop.

14. A 4-bit counter has a maximum modulus of
a) 3
b) 6
c) 8
d) 16
View Answer

Answer: d
Explanation: $2^4 = 16$.

15. A principle regarding most display decoders is that when the correct input is present, the related output will switch
a) HIGH
b) To high impedance
c) To an open
d) LOW
View Answer

Answer: d
Explanation: A principle regarding most display decoders is that when the correct input is present, the related output will switch LOW.

Counter ICs - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Counter ICs”.

1. What is the difference between a 7490 and a 7493?
a) 7490 is a MOD-10, 7493 is a MOD-16
b) 7490 is a MOD-16, 7493 is a MOD-10
c) 7490 is a MOD-12, 7493 is a MOD-16
d) 7490 is a MOD-10, 7493 is a MOD-12
View Answer

Answer: a
Explanation: the difference between a 7490 and a 7493 is that 7490 is a MOD-10, 7493 is a MOD-16 counter.

2. How many different states does a 2-bit asynchronous counter have?
a) 1
b) 4
c) 2
d) 8
View Answer

Answer: b
Explanation: 2^2 = 4.

3. A 12 MHz clock frequency is applied to a cascaded counter containing a modulus-5 counter, a modulus-8 counter, and a modulus-10 counter. The lowest output frequency possible is
_____

a) 10 kHz
b) 20 kHz
c) 30 kHz
d) 60 kHz
View Answer

Answer: c
Explanation: Cascaded counter containing a modulus-5 counter, a modulus-8 counter, and a modulus-10 counter. So, 5*8*10=400. Applied clock frequency = 12 MHz; hence, the lowest output frequency possible is 12MHz/400=30 kHz.

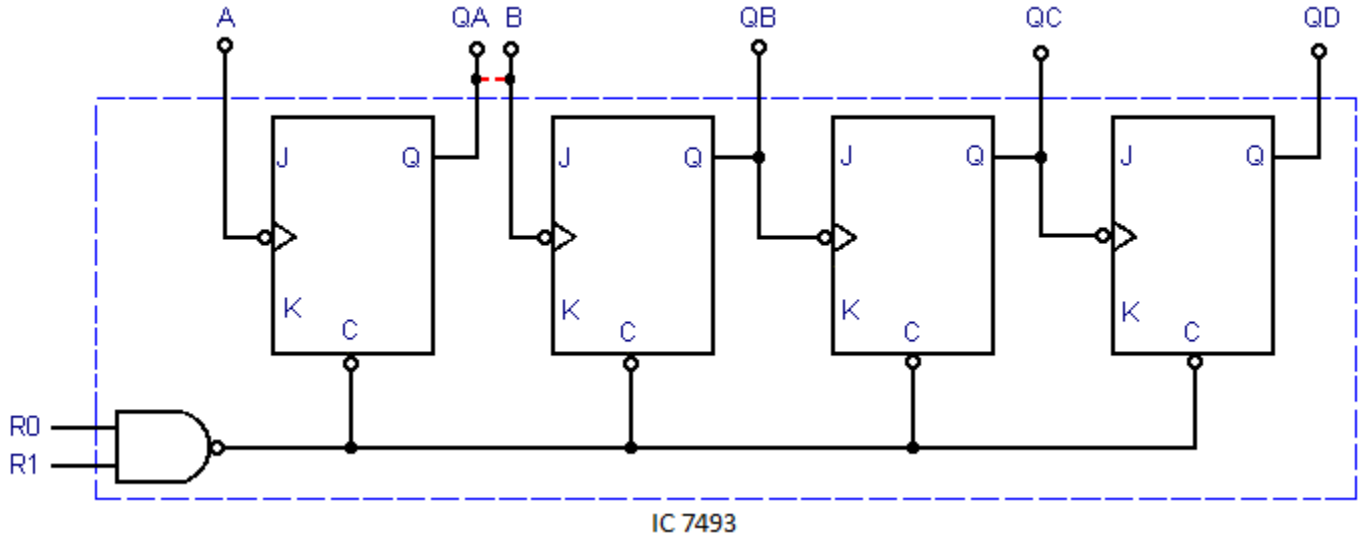
4. Which one is a 4-bit binary ripple counter?
a) IC 7493
b) IC 7490
c) IC 7491
d) IC 7492
View Answer

Answer: b
Explanation: IC 7493 is a 4-bit binary ripple counter.

5. IC 7493 consist of
a) 4 S-R flip-flop
b) 4 J-K flip-flop
c) 4 master-slave flip-flop
d) 4 D flip-flop
View Answer

Answer: c

Explanation: IC 7493 consist of 4 master-slave flip-flop.



6. A reset input is used in IC 7493, why?
a) For increment of bit by 1
b) For decrement of bit by 1
c) For reset the counter
d) None of the Mentioned
View Answer

Answer: c
Explanation: The reset inputs are used to reset the counter to 0000.

7. In a 4-bit binary ripple counter, four master-slave flip-flops are internally connected to provide a _____ bit counter.
a) Divide-by-2 & divide-by-6
b) Divide-by-6 & divide-by-8
c) Divide-by-2 & divide-by-8
d) Divide-by-4 & divide-by-8
View Answer

Answer: c
Explanation: In a 4-bit binary ripple counter, four master-slave flip-flops are internally connected to provide a Divide-by-2 & divide-by-8 bit counter.

8. Which of the following is decade counter?
a) IC 7493
b) IC 7490
c) IC 7491
d) IC 7492
View Answer

Answer: b
Explanation: IC 7490 is called as decade counter or MOD-10.

9. In a 4-bit decade counter, four master-slave flip-flops are internally connected to provide a _____ bit counter.
a) Divide-by-2 & divide-by-6
b) Divide-by-6 & divide-by-8
c) Divide-by-2 & divide-by-5
d) Divide-by-4 & divide-by-8
View Answer

Answer: c
Explanation: In a decade counter, four master-slave flip-flops are internally connected to provide a Divide-by-2 & divide-by-5 bit counter.

10. Reset inputs are used in IC 7490, why?
a) For increment of bit by 1
b) For decrement of bit by 1
c) For reset the counter
d) None of the Mentioned
View Answer

Answer: c
Explanation: The reset inputs are used to reset the counter to 0000.

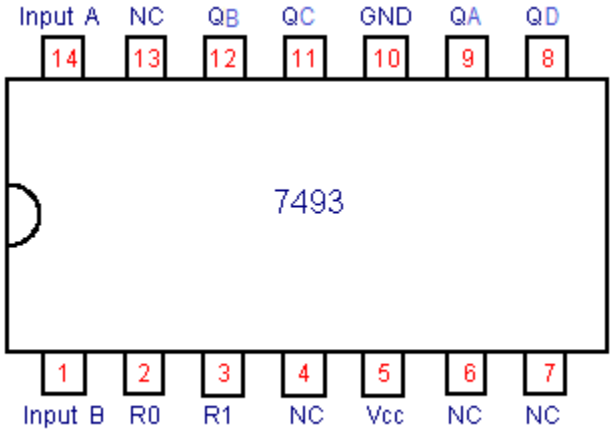
11. The set inputs are used in a decade counter, why?
a) To set the counter to 0011
b) To set the counter to 1000
c) To set the counter to 1001
d) To set the counter to 0001
View Answer

Answer: c
Explanation: The set inputs are used in a decade counter to set set the counter to 1001.

12. List which pins need to be connected together on a 7493 to make a MOD-12 counter.
a) 12 to 1, 11 to 3, 9 to 2
b) 12 to 1, 11 to 3, 12 to 2
c) 12 to 1, 11 to 3, 8 to 2
d) 12 to 1, 11 to 3, 1 to 2
View Answer

Answer: c

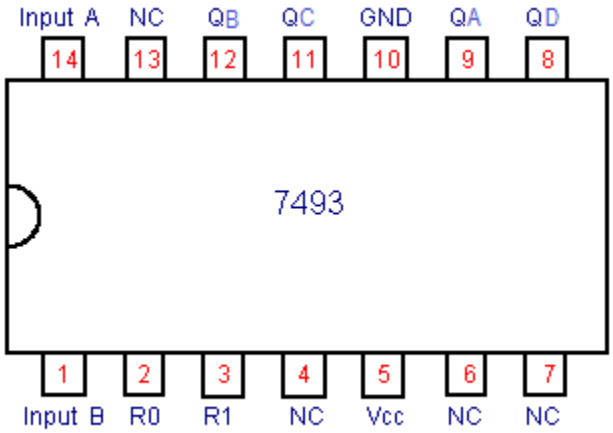
Explanation: It is clear from the diagram shown below: 12 & 1 are clear pins, 11 & 3 are clock pins, 8 & 2 are input for 7493 FF.



13. Ripple counter IC has
a) 10 pins
b) 11 pins
c) 12 pins
d) 14 pins
View Answer

Answer: d

Explanation: Ripple counter IC has 14 pins.



14. Integrated-circuit counter chips are used in numerous applications including:
a) Timing operations, counting operations, sequencing, and frequency multiplication
b) Timing operations, counting operations, sequencing, and frequency division
c) Timing operations, decoding operations, sequencing, and frequency multiplication

d) Data generation, counting operations, sequencing, and frequency multiplication
View Answer

Answer: b
Explanation: There is no integrated Circuits employed for frequency multiplication. In the options a, c, d we have given frequency multiplication. So, they are not the correct answers.

15. What is the difference between a 7490 and a 7492?
a) 7490 is a MOD-12, 7492 is a MOD-10
b) 7490 is a MOD-12, 7492 is a MOD-16
c) 7490 is a MOD-16, 7492 is a MOD-10
d) 7490 is a MOD-10, 7492 is a MOD-12
View Answer

Answer: d
Explanation: From the properties of both ICs, we have 7490 is a MOD-10, 7492 is a MOD-12.

Asynchronous Down Counter - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Asynchronous down counter”.

1. Which of the following statements are true?
a) Asynchronous events does not occur at the same time
b) Asynchronous events are controlled by a clock
c) Synchronous events does not need a clock to control them
d) Only asynchronous events need a control clock
View Answer

Answer: a
Explanation: Asynchronous events does not occur at the same time because of propagation delay.

2. A down counter using n-flip-flops count
a) Downward from a maximum count
b) Upward from a minimum count
c) Downward from a minimum to maximum count
d) None of the Mentioned
View Answer

Answer: a
Explanation: As the name suggests down counter means counting occurs from a higher value to lower value (i.e. (2^n – 1) to 0).

3. UP Counter is
a) It counts in upward manner
b) It count in down ward manner
c) It counts in both the direction
d) None of the mentioned
View Answer

Answer: a
Explanation: UP counter counts in upward manner from 0 to (2^n – 1).

4. DOWN counter is
a) It counts in upward manner
b) It count in downward manner
c) It counts in both the direction
d) None of the mentioned
View Answer

Answer: b
Explanation: DOWN counter counts in downward manner from (2^n – 1) to 0.

5. How many different states does a 3-bit asynchronous down counter have?
a) 2
b) 4
c) 6
d) 8
View Answer

Answer: d
Explanation: The state of a counter doesn't change on changing the direction of count. So, it will have 2^3 = 8 states.

6. In a down counter, which flip-flop doesn't toggle when the inverted output of the preceeding flip-flop goes from HIGH to LOW.
a) MSB flip-flop
b) LSB flip-flop
c) Master slave flip-flop
d) None of the Mentioned
View Answer

Answer: b
Explanation: Since, the LSB flip-flop changes its state at each negative transition of clock. That is why LSB flip-flop doesn't have toggle.

7. In a 3-bit asynchronous down counter, the initial content is
a) 000
b) 111

c) 010
d) 101
View Answer

Answer: a
Explanation: Initially, all the flip-flops are RESET. So, the initial content is 000.

8. In a 3-bit asynchronous down counter, at the first negative transition of the clock, the counter content becomes
a) 000
b) 111
c) 101
d) 010
View Answer

Answer: b
Explanation: Since, in the down counter, the counter content is decremented by 1 for every negative transition. Hence, in a 3-bit asynchronous down counter, at the first negative transition of the clock, the counter content becomes 111.

9. In a 3-bit asynchronous down counter, at the first negative transition of the clock, the counter content becomes
a) 000
b) 111
c) 101
d) 010
View Answer

Answer: c
Explanation: Since, in the down counter, the counter content is decremented by 1 for every negative transition. Hence, in a 3-bit asynchronous down counter, at the first negative transition of the clock, the counter content becomes 101.

10. The hexadecimal equivalent of 15,536 is _____
a) 3CB0
b) 3C66
c) 63C0
d) 6300
View Answer

Answer: a
Explanation: You just divide the number by 2 at the end and use LSB to MSB during arranging them in sequence. And make the pair of four bit from right to left.

11. In order to check the CLR function of a counter
a) Apply the active level to the CLR input and check all of the Q outputs to see if they are all in

their reset state
b) Ground the CLR input and check to be sure that all of the Q outputs are LOW
c) Connect the CLR input to Vcc and check to see if all of the Q outputs are HIGH
d) Connect the CLR to its correct active level while clocking the counter; check to make sure that all of the Q outputs are toggling
View Answer

Answer: a
Explanation: In order to check the CLR function of a counter, apply the active level to the CLR input and check all of the Q outputs to see if they are all in their reset state.

Ripple Counter Propagation Delay Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Propagation Delay in Ripple Counter”.

1. Modulus refers to
a) A method used to fabricate decade counter units
b) The modulus of elasticity, or the ability of a circuit to be stretched from one mode to another
c) An input on a counter that is used to set the counter state, such as UP/DOWN
d) The maximum number of states in a counter sequence
View Answer

Answer: d
Explanation: Modulus is defined as the maximum number of stages/states a counter has.

2. A sequential circuit design is used to
a) Count up
b) Count down
c) Decode an end count
d) Count in a random order
View Answer

Answer: d
Explanation: A sequential circuit design is used to count in a random manner which is faster than combinational circuit.

3. In general, when using a scope to troubleshoot digital systems, the instrument should be triggered by
a) The A channel or channel 1
b) The vertical input mode, when using more than one channel
c) The system clock
d) Line sync, in order to observe troublesome power line glitches
View Answer

Answer: c
Explanation: All the information is sent from one end to another end through the clock pulse which behaves like a carrier. So, for troubleshooting it should be triggered by the same.

4. Which counters are often used whenever pulses are to be counted and the results displayed in decimal?
a) Synchronous
b) Bean
c) Decade
d) BCD
View Answer

Answer: d
Explanation: BCD means Binary Coded Decimal, which means that decimal numbers coded of binary numbers.

5. The _____ counter in the Altera library has controls that allow it to count up or down, and perform synchronous parallel load and asynchronous cascading.
a) 74134
b) LPM
c) Synchronous
d) AHDL
View Answer

Answer: b
Explanation: The library of parameterized modules (LPM) counter in the Altera library has controls that allow it to count up or down, and perform synchronous parallel load and asynchronous cascading.

6. The minimum number of flip-flops that can be used to construct a modulus-5 counter is
a) 3
b) 8
c) 5
d) 10
View Answer

Answer: a
Explanation: The minimum number of flip-flops used in a counter is given by: $2^{(n-1)}<=N<=2^n$.

7. The duty cycle of the most significant bit from a 4-bit (0–9) BCD counter is
a) 20%
b) 50%
c) 10%
d) 80%
View Answer

Answer: a
Explanation: There are 10 states, out of which MSB is high only for (1000, 1001) 2 times. Hence duty cycle is 2/10*100 = 20%.

8. Normally, synchronous counter is designed using
a) S-R flip-flops
b) J-K flip-flops
c) D flip-flops
d) T flip-flops
View Answer

Answer: b
Explanation: Since, J-K flip-flops have options of recovery from toggle condition and by using less number of J-K flip-flops a synchronous counter can be designed. So, it is more preferred.

9. MOD-16 counter requires _____ no. of states.
a) 8
b) 4
c) 16
d) 32
View Answer

Answer: c
Explanation: $2^n >= 2^{(n-1)}$, by using this formula we get the value of N=16.

10. What is state diagram?
a) It provides the graphical representation of states
b) It provides exactly the same information as the state table
c) It is same as the truth table
d) None of the Mentioned
View Answer

Answer: b
Explanation: The state diagram provides exactly the same information as the state table and is obtained directly from the state table.

11. High speed counter is
a) Ring counter
b) Ripple counter
c) Synchronous counter
d) Asynchronous counter
View Answer

Answer: c
Explanation: Synchronous counter doesn't have propagation delay.

12. Program counter in a digital computer
a) Counts the number of programs run in the machine
b) Counts the number of times a subroutine
c) Counts the number of time the loops are executed
d) Points the memory address of the current or the next instruction
View Answer

Answer: d
Explanation: Program counter in a digital computer points the memory address of the current or the next instruction.

13. Fundamental mode is another name for
a) Level operation
b) Pulse operation
c) Clock operation
d) None of the Mentioned
View Answer

Answer: b
Explanation: Whatever the input given to the devices are in the form of pulses always. That is why, it is known as fundamental mode.

Counter Implementation & Applications Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “1’s,2’s,9’s & 10’s Complements”.

1. A ripple counter’s speed is limited by the propagation delay of:
a) Each flip-flop
b) All flip-flops and gates
c) The flip-flops only with gates
d) Only circuit gates
View Answer

Answer: a
Explanation: A ripple counter is something that is derived by other flip-flops. Its like a series of Flip Flops. Output of one FF becomes the input of the next. Because ripple counter is composed of FF only and no gates are there other than FF, so only propagation delay of FF will be taken into account.

2. A 5-bit asynchronous binary counter is made up of five flip-flops, each with a 12 ns propagation delay. The total propagation delay (tp(tot)) is _____
a) 12 ms

b) 24 ns
c) 48 ns
d) 60 ns
View Answer

Answer: d
Explanation: Each bit = 12ns. 5 bits = 12ns * 5 = 60ns.

3. A 4-bit ripple counter consists of flip-flops, which each have a propagation delay from clock to Q output of 15 ns. For the counter to recycle from 1111 to 0000, it takes a total of _____
a) 15 ns
b) 30 ns
c) 45 ns
d) 60 ns
View Answer

Answer: d
Explanation: The given counter is ripple(asynchronous), so only it would take 4 * 15 = 60ns.

4. A ripple counter's speed is limited by the propagation delay of:
a) Each flip-flop
b) All flip-flops and gates
c) The flip-flops only with gates
d) Only circuit gates
View Answer

Answer: a
Explanation: A ripple counter's speed is limited by the propagation delay of each flip-flop.

5. What is the maximum delay that can occur if four flip-flops are connected as a ripple counter and each flip-flop has propagation delays of tPHL = 22 ns and tPLH = 15 ns?
a) 15 ns
b) 22 ns
c) 60 ns
d) 88 ns
View Answer

Answer: d
Explanation: Maximum propagation delay is the longest delay between an input changing value and the output changing value. Hence, 22 * 4 = 88.

6. The main drawback of a ripple counter is that
a) It has a cumulative settling time
b) It has a distributive settling time
c) It has a productive settling time

d) None of the Mentioned
View Answer

Answer: a
Explanation: The main drawback of a ripple counter is that it has a cumulative settling time (i.e. another bit is transmitted just after one consequently).

7. A 4 bit modulo-16 ripple counter uses JK flip-flops. If the propagation delay of each flip-flop is 50 nsec, the maximum clock frequency that can be used is equal to
a) 20 MHz
b) 10 MHz
c) 5 MHz
d) 4 MHz
View Answer

Answer: c
Explanation: Propagation Delay for one FF is 50ns. For 4 FF = 50 x 4 = 200ns. Clock frequency = 1/200ns = 5 MHz.

8. As the number of flip flops are increased, the total propagation delay of
a) Ripple counter increases but that of synchronous counter remains the same
b) Both ripple and synchronous counters increase
c) Both ripple and synchronous counters remain the same
d) Ripple counter remains the same but that of synchronous counter increases
View Answer

Answer: a
Explanation: In ripple counter the clock pulses are applied to one flip- flop only. Hence, as the number of flip-flops increases the delay increases. In synchronous counter, clock pulses to all flip-flops are applied simultaneously.

9. A reliable method for eliminating decoder spikes is the technique called _____
a) Strobing
b) Feeding
c) Wagging
d) Waving
View Answer

Answer: a
Explanation: A reliable method for eliminating decoder spikes is the technique called strobing.

10. A glitch that appears on the decoded output of a ripple counter is often difficult to see on an oscilloscope because
a) It is a random event
b) It occurs less frequently than the normal decoded output
c) It is very fast

d) All of the Mentioned
View Answer

Answer: d
Explanation: A glitch that appears on the decoded output of a ripple counter is often difficult to see on an oscilloscope because it is a random event and very fast and it occurs less frequently than the normal decoded output.

11. Assume a 4-bit ripple counter has a failure in the second flip-flop such that it “locks up”. The third and fourth stages will
a) Continue to count with correct outputs
b) Continue to count but have incorrect outputs
c) Stop counting
d) Turn into molten silicon
View Answer

Answer: c
Explanation: Because next flip-flop’s input depends on the output of the previous flip-flop.

Up down Counter - Digital Circuits Questions and Answers - Sanfoundry

by Manish

This set of Digital Electronics/Circuits Multiple Choice Questions & Answers (MCQs) focuses on “Up down counter”.

1. UP-DOWN counter is a combination of
a) Latches
b) Flip-flops
c) UP counter
d) Up counter & down counter
View Answer

Answer: d
Explanation: As the name suggests UP-DOWN, it means that it has up-counter and down-counter as well.

2. UP-DOWN counter is also known as
a) Dual counter
b) Multi counter
c) Multimode counter
d) None of the Mentioned
View Answer

Answer: c
Explanation: UP-DOWN counter is also known as multimode counter because it has capability of counting upward as well as downwards.

3. In an UP-counter, each flip-flop is triggered by
a) The output of the next flip-flop
b) The normal output of the preceding flip-flop
c) The clock pulse of the previous flip-flop
d) The inverted output of the preceding flip-flop
View Answer

Answer: b
Explanation: In an UP-counter, each flip-flop is triggered by the normal output of the preceding flip-flop.

4. In DOWN-counter, each flip-flop is triggered by
a) The output of the next flip-flop
b) The normal output of the preceding flip-flop
c) The clock pulse of the previous flip-flop
d) The inverted output of the preceding flip-flop
View Answer

Answer: d
Explanation: In DOWN-counter, each flip-flop is triggered by the inverted output of the preceding flip-flop.

5. Binary counter that count incrementally and decremently is called
a) Up-down counter
b) LSI counters
c) Down counter
d) Up counter
View Answer

Answer: a
Explanation: Binary counter that count incrementally and decremently is called UP-DOWN counter/multimode counter.

6. Once an up-/down-counter begins its count sequence, it
a) Starts counting
b) Can be reversed
c) Can’t be reversed
d) None of the Mentioned
View Answer

Answer: d
Explanation: In up/down ripple counter once the counting begins, we can simply change the pulse M (mode control) M = 0 or 1 respectively for UP counter or Down counter.

7. In 4-bit up-down counter, how many flip-flops are required?
a) 2
b) 3
c) 4
d) 5
View Answer

Answer: c
Explanation: In a 4-bit up-down counter, there are 4 J-K flip-flops required.

8. A modulus-10 counter must have _____
a) 10 flip-flops
b) Flip-flops
c) 2 flip-flops
d) Synchronous clocking
View Answer

Answer: b
Explanation: For any number of modulus we need only 1 flip flop. Because it takes 1 bit but works as a many modulus.

9. Which is not an example of a truncated modulus?
a) 8
b) 9
c) 11
d) 15
View Answer

Answer: a
Explanation: An n-bit counter whose modulus is less than the maximum possible is called a truncated counter.

10. The designation means that the _____
a) Up count is active-HIGH, the down count is active-LOW
b) Up count is active-LOW, the down count is active-HIGH
c) Up and down counts are both active-LOW
d) Up and down counts are both active-HIGH
View Answer

Answer: a
Explanation: The designation means that the up count is active-HIGH, the down count is active-LOW.

11. An asynchronous binary up counter, made from a series of leading edge-triggered flip-flops, can be changed to a down counter by _____
a) Taking the output on the other side of the flip-flops (instead of Q)
b) Clocking of each succeeding flip-flop from the other side (instead of Q)
c) Changing the flip-flops to trailing edge triggering
d) All of the Mentioned
View Answer

Answer: d
Explanation: By all of the mentioned ideas, an asynchronous binary up counter, made from a series of leading edge-triggered flip-flops, can be changed to a down counter.

12. A 4-bit binary up counter has an input clock frequency of 20 kHz. The frequency of the most significant bit is _____
a) 1.25 kHz
b) 2.50 kHz
c) 160 kHz
d) 320 kHz
View Answer

Answer: a
Explanation: Input clock is given by: 20/2 kHz. So, count on basis of 10 kHz clock. And MSB changes on 8th stage; Hence, f = 10/8 = 1.25 kHz.